Logical Operators (Connectives)

- We will examine the following logical operators:

- Negation (NOT)
- Conjunction (AND)
- Disjunction (OR)
- Exclusive or (XOR)
- Implication (if – then)
- Biconditional (if and only if)

- Truth tables can be used to show how these operators can combine propositions to compound propositions.

Negation (NOT)

- Unary Operator, Symbol: $\neg$

P	$\neg P$
true (T)	false (F)
false (F)	true (T)

Conjunction (AND)

- Binary Operator, Symbol: $\wedge$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction (OR)

- Binary Operator, Symbol: $\vee$

P	Q	P∨Q
T	T	T
T	F	T
F	T	T
F	F	F

Implication (if - then)

- Binary Operator, Symbol: $\rightarrow$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional (if and only if)

- Binary Operator, Symbol: $\leftrightarrow$

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Statements and Operators

- Statements and operators can be combined in any way to form new statements.

P	Q	$\neg P$	$\neg Q$	$(\neg P)\vee(\neg Q)$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Statements and Operations

- Statements and operators can be combined in any way to form new statements.

P	Q	$P \wedge Q$	$\neg\,(P \wedge Q)$	$(\neg P) \vee (\neg Q)$
T	T	T	F	F
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

Equivalent Statements

P	Q	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$	$\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$
T	T	F	F	T
T	F	T	T	T
F	T	T	T	T
F	F	T	T	T

- The statements $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$ are logically equivalent, since $\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$ is always true.

TRUTH TABLES FOR COMPOUND PROPOSITIONS

- Construction of a truth table:
- Rows
  - Need a row for every possible combination of values for the atomic propositions.
- Columns
  - Need a column for the compound proposition (usually at far right)
  - Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
    - This includes the atomic propositions

Atomic Statements:

Declarative sentences which cannot be further spilt into simple sentences
are called as atomic statements (or primary statements or primitive statements).
Example: p is a prime number.

EXAMPLE TRUTH TABLE

- Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	¬r	p ∨ q	p ∨ q → ¬r
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Example 1.1 Construct a truth table for each of the following compound propositions:

(a) $(p \vee q) \rightarrow (p \wedge q)$;

(b) $(p \rightarrow q) \rightarrow (q \rightarrow p)$;

(c) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$;

(d) $(p \leftrightarrow q) \leftrightarrow ((p \wedge q) \vee (\neg p \wedge \neg q))$;

(e) $(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$.

(a) **Table 1.15** Truth Table for $(p \vee q) \rightarrow (p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

EQUIVALENT PROPOSITIONS

- Two propositions are *equivalent* if they always have the same truth value.
- **Example**: Show using a truth table that the conditional is equivalent to the contrapositive.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

USING A TRUTH TABLE TO SHOW NON-EQUIVALENCE

Example: Show using truth tables that the converse and inverse of an implication are not equivalent to the implication.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

TAUTOLOGIES, CONTRADICTIONS, AND CONTINGENCIES

- A tautology is a proposition which is always true.
  - Example: $p \vee \neg p$
- A *contradiction* is a proposition which is always false.
  - Example: $p \wedge \neg p$
- A *contingency* is a proposition which is neither a tautology nor a contradiction, such as p

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

LOGICALLY EQUIVALENT

- Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table shows that $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Q1. Prove $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Note We have already noted that the biconditional proposition $A \leftrightarrow B$ is true whenever both A and B have the same truth value, viz. $A \leftrightarrow B$ is a tautology, when A and B are equivalent.

Conversely, $A \equiv B$, when $A \leftrightarrow B$ is a tautology. For example, $(p \to q) \equiv (\neg p \vee q)$, since $(p \to q) \leftrightarrow (\neg p \vee q)$ is a tautology, as seen from the truth Table 1.8 given below:

p	q	$p \to q$	$\neg p$	$\neg p \vee q$	$(p \to q) \leftrightarrow (\neg p \vee q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

DUALITY LAW

The *dual* of a compound proposition that contains only the logical operators $\vee$, $\wedge$ and $\urcorner$ is the proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each T by F and each F by T, where T and F are special variables representing compound propositions that are tautologies and contradictions respectively. The dual of a proposition A is denoted by A^*.

Example. The dual of $(T \wedge p) \vee q$ is $(F \vee p) \wedge q$

DUALITY THEOREM

If $A(p_1, p_2, \ldots, p_n) \equiv B(p_1, p_2, \ldots, p_n)$, where A and B are compound propositions, then $A^*(p_1, p_2, \ldots, p_n) \equiv B^*(p_1, p_2, \ldots, p_n)$.

ALGEBRA OF PROPOSITIONS

A proposition in a compound proposition can be replaced by one that is equivalent to it without changing the truth value of the compound proposition. By this way, we can construct new equivalences (or laws).

For example, we have proved that $p \to q \equiv \neg p \vee q$

Table 1.9 Laws of Algebra of Propositions

Sl. No.	*Name of the law*	*Primal form*	*Dual form*
1.	Idempotent law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2.	Identity law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3.	Dominant law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4.	Complement law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5.	Commutative law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6.	Associative law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7.	Distributive law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8.	Absorption law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9.	De Morgan's law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$

Table 1.10 Equivalences Involving Conditionals

1. $p \to q \equiv \neg p \vee q$
2. $p \to q \equiv \neg q \to \neg p$
3. $p \vee q \equiv \neg p \to q$
4. $p \wedge q \equiv \neg(p \to \neg q)$
5. $\neg(p \to q) \equiv p \wedge \neg q$
6. $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$
7. $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$
8. $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$
9. $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

Table 1.11 Equivalences Involving Biconditionals

1. $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
2. $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
3. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
4. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

TAUTOLOGICAL IMPLICATION

A compound proposition $A(p_1, p_2, \ldots, p_n)$ is said to *tautologically imply* or simply *imply* the compound proposition $B(p_1, p_2, \ldots, p_n)$, if B is true whenever A is true or equivalently if and only if $A \rightarrow B$ is a tautology. This is denoted by $A \Rightarrow B$, read as "A implies B".

For example, $p \Rightarrow p \vee q$, as seen from the following truth Table 1.12. We note that $p \vee q$ is true, whenever p is true and that $p \rightarrow (p \vee q)$ is a tautology.

Table 1.12

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$	$(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

$$(p \rightarrow q) \Rightarrow (\neg q \rightarrow \neg p)$$

Some important implications which can be proved by truth tables are given in Table 1.14.

Table 1.14 Implications

1. $p \wedge q \Rightarrow p$
2. $p \wedge q \Rightarrow q$
3. $p \Rightarrow p \vee q$
4. $\neg p \Rightarrow p \rightarrow q$
5. $q \Rightarrow p \rightarrow q$
6. $\neg(p \rightarrow q) \Rightarrow p$
7. $\neg(p \rightarrow q) \Rightarrow \neg q$
8. $p \wedge (p \rightarrow q) \Rightarrow q$
9. $\neg q \wedge (p \rightarrow q) \Rightarrow \neg p$
10. $\neg p \wedge (p \vee q) \Rightarrow q$
11. $(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r$
12. $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r) \Rightarrow r$

Note We can easily verify that if $A \Rightarrow B$ and $B \Rightarrow A$, then $A \equiv B$. Hence to prove the equivalence of two propositions, it is enough to prove that each implies the other.

EQUIVALENCE PROOFS

Example: Show that $\neg(p \vee (\neg p \wedge q))$

is logically equivalent to $\neg p \wedge \neg q$

Solution:

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{De Morgan law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{De Morgan law} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{double negation law} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{distributive law} \\
&\equiv F \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv F \\
&\equiv (\neg p \wedge \neg q) \vee F && \text{commutative law} \\
&\equiv (\neg p \wedge \neg q) && \text{identity law for } \mathbf{F}
\end{aligned}
$$

EQUIVALENCE PROOFS

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Solution:

$$
\begin{aligned}
(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{by truth table for } \rightarrow \\
&\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{by the first De Morgan law} \\
&\equiv (\neg p \vee p) \vee (\neg q \vee q) && \text{by associative and} \\
& && \text{commutative laws} \\
& && \text{laws for disjunction} \\
&\equiv T \vee T && \text{by truth tables} \\
&\equiv T && \text{by the domination law}
\end{aligned}
$$

(i) Show that $(p \vee q) \wedge \neg p \Leftrightarrow (\neg p \wedge q)$

$S \Leftrightarrow (p \vee q) \wedge \neg p$	Reasons
$\Leftrightarrow (p \vee q) \wedge \neg p$	Given
$\Leftrightarrow (p \wedge \neg p) \vee (q \wedge \neg p)$	Distributive law
$\Leftrightarrow F \vee (\neg p \wedge q)$	Negation law,Commutative law
$\Leftrightarrow \neg p \wedge q$	Identity law

1.	Idempotent law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2.	Identity law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3.	Dominant law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4.	Complement law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5.	Commutative law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6.	Associative law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7.	Distributive law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8.	Absorption law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9.	De Morgan's law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$

(ii) Show that $(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$

$S \Leftrightarrow (p \vee q) \wedge \neg(\neg p \wedge q)$	Reasons
$\Leftrightarrow (p \vee q) \wedge \neg(\neg p \wedge q)$	Given
$\Leftrightarrow (p \vee q) \wedge (\neg\neg p \vee \neg q)$	De Morgan's law
$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q)$	Negation law
$\Leftrightarrow p \vee (q \wedge \neg q)$	Distributive law
$\Leftrightarrow p \vee F$	Negation law
$\Leftrightarrow p$	Identity law

1.	Idempotent law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2.	Identity law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3.	Dominant law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4.	Complement law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5.	Commutative law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6.	Associative law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7.	Distributive law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8.	Absorption law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9.	De Morgan's law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$

(iii) Show that $\neg(p \vee (\neg p \wedge q)) \Leftrightarrow \neg p \wedge \neg q$

$S \Leftrightarrow \neg(p \vee (\neg p \wedge q))$	Reasons
$\Leftrightarrow \neg(p \vee (\neg p \wedge q))$	Given
$\Leftrightarrow \neg p \wedge \neg(\neg p \wedge q))$	De Morgan's law
$\Leftrightarrow \neg p \wedge (p \vee \neg q))$	De Morgan's law
$\Leftrightarrow (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	Distributive law
$\Leftrightarrow F \vee (\neg p \wedge \neg q)$	Negation law
$\Leftrightarrow \neg p \wedge \neg q$	Identity law

1.	Idempotent law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2.	Identity law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3.	Dominant law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4.	Complement law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5.	Commutative law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6.	Associative law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7.	Distributive law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8.	Absorption law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9.	De Morgan's law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$

(iv) Show that $\neg(\neg((p \vee q) \wedge r) \vee \neg q) \Leftrightarrow q \wedge r$

$S \Leftrightarrow \neg(\neg((p \vee q) \wedge r) \vee \neg q) \Leftrightarrow q \wedge r$	Reasons
$\Leftrightarrow \neg(\neg((p \vee q) \wedge r) \vee \neg q)$	Given
$\Leftrightarrow ((p \vee q) \wedge r) \wedge q$	De Morgan's law
$\Leftrightarrow (p \vee q) \wedge (r \wedge q)$	Associative law
$\Leftrightarrow (p \wedge (r \wedge q)) \vee (q \wedge (r \wedge q))$	Distributive law
$\Leftrightarrow (p \wedge (r \wedge q)) \vee (r \wedge q)$	Idempotent law
$\Leftrightarrow r \wedge q$	Absorption law

1.	Idempotent law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2.	Identity law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3.	Dominant law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4.	Complement law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5.	Commutative law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6.	Associative law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7.	Distributive law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8.	Absorption law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9.	De Morgan's law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$

1. Examples on Truth table construction for compound proposition

- Construct the truth table for each of the compound propositions given as follows:
  - $(p \to (q \to s)) \wedge (\urcorner r \vee p) \wedge q$
  - $((p \to q) \to r) \to s$

Truth Table for $(p \rightarrow (q \rightarrow s)) \wedge (\neg r \vee p) \wedge q$

p	q	r	s	$q \rightarrow s \equiv a$	$p \rightarrow a \equiv b$	$\neg r$	$(\neg r \vee p) \equiv c$	$b \wedge c$	$b \wedge c \wedge q$
T	T	T	T	T	T	F	T	T	T
T	T	T	F	F	F	F	T	F	F
T	T	F	T	T	T	T	T	T	T
T	T	F	F	F	F	T	T	F	F
T	F	T	T	T	T	F	T	T	F
T	F	T	F	T	T	F	T	T	F
T	F	F	T	T	T	T	T	T	F
T	F	F	F	T	T	T	T	T	F
F	T	T	T	T	T	F	F	F	F
F	T	T	F	F	T	F	F	F	F
F	T	F	T	T	T	T	T	T	T
F	T	F	F	F	T	T	T	T	T
F	F	T	T	T	T	F	F	F	F
F	F	T	F	T	T	F	F	F	F
F	F	F	T	T	T	T	T	T	F
F	F	F	F	T	T	T	T	T	F

Truth Table for $((p \to q) \to r) \to s$

p	q	r	s	$p \to q$	$(p \to q) \to r$	$((p \to q) \to r) \to s$
T	T	T	T	T	T	T
T	T	T	F	T	T	F
T	T	F	T	T	F	T
T	T	F	F	T	F	T
T	F	T	T	F	T	T
T	F	T	F	F	T	F
T	F	F	T	F	T	T
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	T	F	T	T	F
F	T	F	T	T	F	T
F	T	F	F	T	F	T
F	F	T	T	T	T	T
F	F	T	F	T	T	F
F	F	F	T	T	F	T
F	F	F	F	T	F	T

2. Examples on Tautologies, Contradictions identifying using truth tables.

- Determine which of the following compound propositions are tautologies and which of them are contradictions, using truth tables:
  - $\neg(q \to r) \wedge r \wedge (p \to q)$
  - $((p \to q) \wedge (q \to r)) \to (p \to r)$

Truth Tables for $\urcorner(q \to r) \wedge r \wedge (p \to q)$

p	q	r	$p \to q$	$q \to r$	$\urcorner(q \to r)$	$\urcorner(q \to r) \wedge r$	$\urcorner(q \to r) \wedge r \wedge (p \to q)$
T	T	T	T	T	F	F	F
T	T	F	T	F	T	F	F
T	F	T	F	T	F	F	F
T	F	F	F	T	F	F	F
F	T	T	T	T	F	F	F
F	T	F	T	F	T	F	F
F	F	T	T	T	F	F	F
F	F	F	T	T	F	F	F

The last column contains only F as the truth values of the given statement. Hence it is a contradiction.

Truth Table for $((p \to q) \wedge (q \to r)) \to (p \to r)$

p	q	r	$p \to q$	$p \to r$	$q \to r$	$(p \to q) \wedge (q \to r)$	$((p \to q) \wedge (q - r)) \to (p - r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	F	T
T	F	F	F	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

3. Examples on Equivalence and Tautology Proof without using truth tables.

- Without using truth tables, prove the following.

1. $(\neg p \vee q) \wedge (p \wedge (p \wedge q) \equiv p \wedge q$
2. $p \rightarrow (q \rightarrow p) \equiv \neg p \rightarrow (p \rightarrow q)$
3. $((p \vee q) \wedge \neg(\neg p \wedge (\neg q \vee \neg r))) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$ is a tautology.

Proof 1.

$$
\begin{aligned}
(\neg p \vee q) \wedge (p \wedge (p \wedge q)) &\equiv (\neg p \vee q) \wedge (p \wedge p) \wedge q, \text{ by associative law} \\
&\equiv (\neg p \vee q) \wedge (p \wedge q), \text{ by idempotent law} \\
&\equiv (p \wedge q) \wedge (\neg p \vee q), \text{ by commutative law} \\
&\equiv ((p \wedge q) \wedge \neg p) \vee ((p \wedge q) \wedge q), \text{ by distributive law} \\
&\equiv (\neg p \wedge (p \wedge q)) \vee ((p \wedge q) \wedge q), \text{ by commutative law} \\
&\equiv ((\neg p \wedge p) \wedge q) \vee (p \wedge (q \wedge q), \text{ by associative law} \\
&\equiv (F \vee q) \vee (p \wedge q), \text{ by complement and idempotent law} \\
&\equiv F \vee (p \wedge q), \text{ by dominant law} \\
&\equiv p \wedge q, \text{ by dominant law.}
\end{aligned}
$$

Proof 2.

$$p \to (q \to p) \equiv \neg p \vee (q \to p) \text{ by (1) of Table}$$
$$\equiv \neg p \vee (\neg q \vee p) \text{ by (1) of Table}$$
$$\equiv \neg q \vee (p \vee \neg p), \text{ by commutative and associative laws}$$
$$\equiv \neg p \vee \mathrm{T}, \text{ by complement law}$$
$$\equiv \mathrm{T}, \text{ by dominant law}$$

$$\neg p \to (p \to q) \equiv p \vee (p \to q), \text{ by (1) of Table}$$
$$\equiv p \vee (\neg p \vee q), \text{ by (1) of Table}$$
$$\equiv (p \vee \neg p) \vee q, \text{ by associative l}$$
$$\equiv T \vee q, \text{ by complement law}$$
$$\equiv T, \text{ by dominant law,}$$

From (1) and (2), the result follows.

Equivalences Involving Conditionals

1.	$p \to q \equiv \neg p \vee q$
2.	$p \to q \equiv \neg q \to \neg p$
3.	$p \vee q \equiv \neg p \to q$
4.	$p \wedge q \equiv \neg(p \to \neg q)$
5.	$\neg(p \to q) \equiv p \wedge \neg q$
6.	$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$
7.	$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$
8.	$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$
9.	$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

Proof 3.

$$((p \vee q) \wedge \urcorner(\urcorner p \wedge (\urcorner q \vee \urcorner r))) \vee (\urcorner p \wedge \urcorner q) \vee (\urcorner p \wedge \urcorner r)$$

$$\equiv ((p \vee q) \wedge \urcorner(\urcorner p \wedge \urcorner(q \wedge r))) \vee \urcorner(p \vee q) \vee \urcorner(p \vee r),$$

by De Morgan's law

$$\equiv ((p \vee q) \wedge (p \vee (q \wedge r))) \vee \urcorner(p \vee q) \vee \urcorner(p \vee r),$$

by De Morgan's law

$$\equiv ((p \vee q) \wedge [(p \vee q) \wedge (p \vee r)]) \vee [\urcorner(p \vee q) \vee \urcorner(p \vee r)],$$

by distributive law

$$\equiv [(p \vee q) \wedge (p \vee r)] \vee \urcorner[(p \vee q) \wedge (p \vee r)],$$

by idempotent and De Morgan's laws

The final statement is in the form of $p \vee \urcorner p$.

$\therefore$ L.H.S. $\equiv$ T

Hence the given statement is tautology.

4. Examples on Equivalence Proof using Dual.

- Prove the following equivalences by proving the equivalences of the duals.

1. $\neg((\neg p \wedge q) \vee (\neg p \wedge \neg q)) \vee (p \wedge q) \equiv p$
2. $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \wedge r)$

Proof 1.

The dual of the given equivalence is

$$\neg((\neg p \vee q) \wedge (\neg p \vee \neg q)) \wedge (p \vee q) \equiv p$$

Let us now prove the dual equivalence.

$$\begin{aligned}
\text{L.H.S.} &\equiv \neg(\neg p \vee (q \wedge \neg q)) \wedge (p \vee q), \text{ by distribution law} \\
&\equiv \neg(\neg p \vee F) \wedge (p \vee q), \text{ by complement law} \\
&\equiv \neg(\neg p) \wedge (p \vee q), \text{ by identity law} \\
&\equiv p \wedge (p \vee q) \\
&\equiv p, \text{ by absorption law}
\end{aligned}$$

Proof 2.

$(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

i.e., $\urcorner(p \vee q) \vee r \equiv (\urcorner p \vee r) \wedge (\urcorner q \vee r)$

Dual of this equivalence is

$$\urcorner(p \wedge q) \wedge r \equiv (\urcorner p \wedge r) \vee (\urcorner q \wedge r)$$

L.H.S $\equiv (\urcorner p \vee \urcorner q) \wedge r$, by De Morgan's law

$\equiv (\urcorner p \wedge r) \vee (\urcorner q \wedge r)$, by distributive law

$\equiv$ R.H.S.

5. Examples on Proof of Implications using truth tables.

- Prove the following implications by using truth tables.

1. $(p \to (q \to r)) \Longrightarrow (p \to q) \to (p \to r)$

Proof 1.

We have defined that $A \Rightarrow B$, if and only if $A \rightarrow B$ is a tautology

p	q	r	$p \rightarrow q$ (a)	$q \rightarrow r$ (b)	$p \rightarrow r$ (c)	$p \rightarrow b$ (d)	$a \rightarrow c$ (e)	$d \rightarrow e$
T	T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F	T
T	F	T	F	T	T	T	T	T
T	F	F	F	T	F	T	T	T
F	T	T	T	T	T	T	T	T
F	T	F	T	F	T	T	T	T
F	F	T	T	T	T	T	T	T
F	F	F	T	T	T	T	T	T

Since $d \rightarrow e$, viz., $[p \rightarrow (q \rightarrow r)] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$ is a tautology, the required implication follows.

6. Examples on Proof of Implications without using truth tables.

➢ Prove the following implications without using truth tables.

1. $(p \vee q) \wedge (p \to r) \wedge (q \to r) \Rightarrow r$

Proof 1.

$$[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$$

$$\equiv (p \vee q) \wedge ((p \vee q) \rightarrow r) \rightarrow r, \text{ from Table}$$

$$\equiv (p \vee q) \wedge (\neg(p \vee q) \vee r) \rightarrow r$$

$$\equiv (F \vee (p \vee q) \wedge r) \rightarrow r$$

$$\equiv ((p \vee q) \wedge r) \rightarrow r$$

$$\equiv \neg((p \vee q) \wedge r) \vee r$$

$$\equiv \neg((p \wedge r) \vee (q \wedge r)) \vee r$$

$$\equiv (\neg(p \wedge r) \wedge \neg(q \wedge r)) \vee r$$

$$\equiv (\neg(p \wedge r) \vee r) \wedge (\neg(q \wedge r) \vee r)$$

$$\equiv (\neg p \vee \neg r \vee r) \wedge (\neg q \vee \neg r \vee r)$$

$$\equiv (\neg p \vee T) \wedge (\neg q \vee T)$$

$$\equiv T \wedge T$$

$$\equiv T$$

1. $p \rightarrow q \equiv \neg p \vee q$
2. $p \rightarrow q \equiv \neg q \rightarrow \neg p$
3. $p \vee q \equiv \neg p \rightarrow q$
4. $p \wedge q \equiv \neg(p \rightarrow \neg q)$
5. $\neg(p \rightarrow q) \equiv p \wedge \neg q$
6. $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
7. $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
8. $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
9. $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Normal Forms

- If a given statement formula $A(p_1, p_2, \ldots p_n)$ involves n atomic variables, we have 2^n possible combinations of truth values of statements replacing the variables.
- The problem of determining whether a given statement formula is a Tautology, or a Contradiction is called a decision problem.
- The construction of truth table involves a finite number of steps, but the construction may not be practical.
- We therefore reduce the given statement formula to normal form and find whether a given statement formula is a Tautology or Contradiction.
- There are two types of normal form—disjunctive normal form (DNF) and conjunctive normal form (CNF).
- It will be convenient to use the word **product** in place of **conjunction** and **sum** in place of **disjunction** in our current discussion.

Basic Terminologies:

- Elementary product: A product of the variables and their negations (*a conjunction of primary statements and their negations*) is called an elementary product.
- Elementary Sum: A sum of the variables and their negations is called an elementary sum.

For example, p, $\urcorner p$, $p \wedge \urcorner p$, $\urcorner p \wedge q$, $p \wedge \urcorner q$ and $\urcorner p \wedge \urcorner q$ some elementary products in 2 variables, q, $\urcorner q$, $p \vee q$, $p \vee \urcorner q$ and $\urcorner p \vee \urcorner q$ are some elementary sums is 2 variables.

❖ Disjunctive Normal Form: A compound proposition (or a formula) which consists of a sum of **elementary products** and which is equivalent to a given proposition is called a disjunctive normal form (DNF) of the given proposition.

For example, the DNF of $q \to (q \to p)$ is given by

$$\begin{aligned} q \to (q \to p) &\equiv \neg q \vee (q \to p) \\ &\equiv \neg q \vee (\neg q \vee p) \\ &\equiv (\neg q \vee \neg q) \vee p, \text{ by associative law} \\ &\equiv \neg q \vee p, \text{ by idempotent law.} \end{aligned}$$

❖ Conjunctive Normal Form: A formula which consists of a product of **elementary sums** and which is equivalent to a given formula is called a conjunctive normal form (CNF) of the given formula.

The CNF of $\neg(p \vee q) \leftrightarrow (p \wedge q)$ is given by

$$\begin{aligned}
\neg(p \vee q) \leftrightarrow (p \wedge q) &\equiv (\neg(p \vee q) \wedge (p \wedge q)) \vee (\neg(\neg(p \vee q)) \wedge \neg(p \wedge q)) \\
&\equiv (\neg p \wedge \neg q) \wedge (p \wedge q) \vee (p \vee q) \wedge (\neg p \vee \neg q) \\
&\equiv (p \wedge \neg p) \wedge (q \wedge \neg q) \vee (p \vee q) \wedge (\neg p \vee \neg q) \\
&\equiv F \wedge F \vee (p \vee q) \wedge (\neg p \vee \neg q) \\
&\equiv (p \vee q) \wedge (\neg p \vee \neg q)
\end{aligned}$$

Table 1.11 Equivalences Involving Biconditionals

1. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
2. $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
3. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
4. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Procedure to Obtain the DNF or CNF of a Given Formula

Step 1

If the connectives $\to$ and $\leftrightarrow$ are present in the given formula they are replaced by $\wedge$, $\vee$ and $\urcorner$ viz. $p \to q$ is replaced by $\urcorner p \vee q$ and $p \leftrightarrow q$ is replaced by either $(p \wedge q) \vee (\urcorner p \wedge \urcorner q)$ or $(\urcorner p \vee q) \wedge (\urcorner q \vee p)$.

Step 2

If the negation is present before the given formula or a part of the given formula (not a variable), De Morgan's laws are applied so that the negation is brought before the variables only.

Step 3

If necessary, the distributive law and the idempotent law are applied.

Step 4

If there is an elementary product which is equivalent to the truth value F in the DNF, it is omitted. Similarly if there is an elementary sum which is equivalent to the truth value T in the CNF, it is omitted.

- Find the disjunctive normal forms of the following statements:

(i) $\neg(\neg(p \leftrightarrow q) \wedge r)$

(ii) $p \vee (\neg p \rightarrow (q \vee (q \rightarrow \neg r)))$

(iii) $p \wedge \neg(q \wedge r) \vee (p \rightarrow q)$

$$
\begin{aligned}
\text{(i)}\quad \neg(\neg(p \leftrightarrow q) \wedge r) &\equiv \neg(\neg((p \wedge q) \vee (\neg p \wedge \neg q)) \wedge r) \\
&\equiv \neg[(\neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q)) \wedge r] \\
&\equiv \neg[((\neg p \vee \neg q) \wedge (p \vee q)) \wedge r] \\
&\equiv \neg[((\neg p \wedge p) \vee (\neg p \wedge q) \vee (\neg q \wedge p) \vee (\neg q \wedge q)) \wedge r], \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{by extended distributive law} \\
&\equiv \neg[((\neg p \wedge q) \vee (\neg q \wedge p)) \wedge r) \\
&\equiv \neg[((\neg p \vee \neg q) \wedge (\neg p \vee p) \wedge (q \vee \neg q) \wedge (q \vee p)) \wedge r] \\
&\equiv \neg[((p \vee q) \wedge (\neg p \vee \neg q)) \wedge r] \\
&\equiv \neg(p \vee q) \vee \neg(\neg p \vee \neg q) \vee \neg r \\
&\equiv (\neg p \wedge \neg q) \vee (p \wedge q) \vee \neg r
\end{aligned}
$$

$$
\begin{aligned}
\text{(ii)}\quad p \vee (\neg p \rightarrow (q \vee (q \rightarrow \neg r)))
&\equiv p \vee (\neg p \rightarrow (q \vee (\neg q \vee \neg r))) \\
&\equiv p \vee (p \vee (q \vee (\neg q \vee \neg r))) \\
&\equiv p \vee p \vee q \vee \neg q \vee \neg r \\
&\equiv p \vee q \vee \neg q \vee \neg r
\end{aligned}
$$

(iii) $p \wedge \urcorner(q \wedge r) \vee (p \rightarrow q)$

$$\begin{aligned}
&\equiv p \wedge \urcorner(q \wedge r) \vee (\urcorner p \vee q) \\
&\equiv (p \wedge (\urcorner q \vee \urcorner r)) \vee (\urcorner p \vee q) \\
&\equiv (p \wedge \urcorner q) \vee (p \wedge \urcorner r) \vee (\urcorner p \vee q) \\
&\equiv (p \wedge \urcorner q) \vee (p \wedge \urcorner r) \vee \urcorner p \vee q
\end{aligned}$$

- Find the conjunctive normal forms of the following statements:

(i) $(p \wedge \neg(q \wedge r)) \vee (p \to q)$

(ii) $(q \vee (p \wedge q)) \wedge \neg((p \vee r) \wedge q)$

(iii) $(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \vee p)$

(i) $(p \wedge \neg(q \wedge r)) \vee (p \rightarrow q)$

$$\equiv (p \wedge (\neg q \vee \neg r)) \vee (\neg p \vee q)$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \vee q)$$

$$\equiv (p \vee p) \wedge (p \vee \neg r) \wedge (\neg q \vee p) \wedge (\neg q \vee \neg r) \vee (\neg p \vee q)$$

$$\equiv (p \vee p) \wedge (p \vee \neg r) \wedge (p \vee \neg q) \wedge (\neg p \vee q \vee \neg q \vee \neg r)$$

$$\equiv (p \vee p) \wedge (p \vee \neg r) \wedge (p \vee \neg q) \wedge (\neg p \vee T \vee \neg r)$$

$$\equiv (p \wedge (p \vee \neg r) \wedge (p \vee \neg q)$$

(ii) $[q \vee (p \wedge q)] \wedge \neg[(p \vee r) \wedge q]$

$$\equiv q \wedge \neg[(p \vee r) \wedge q], \text{ by absorption law}$$

$$\equiv q \wedge [\neg(p \vee r) \vee \neg q]$$

$$\equiv q \wedge [(\neg p \wedge \neg r) \vee \neg q]$$

$$\equiv q \wedge (\neg p \vee \neg q) \wedge (\neg q \vee \neg r)$$

(iii) $(p \wedge \urcorner(q \vee r)) \vee (((p \wedge q) \vee \urcorner r) \wedge p)$

$$\begin{aligned}
&\equiv (p \wedge (\urcorner q \wedge \urcorner r)) \vee ((p \vee \urcorner r) \wedge (q \vee \urcorner r) \wedge p)\\
&\equiv (p \wedge \urcorner q \wedge \urcorner r) \vee (p \wedge (p \vee \urcorner r) \wedge q \vee \urcorner r))\\
&\equiv (p \wedge \urcorner q \wedge \urcorner r) \vee (p \wedge (q \vee \urcorner r)), \text{ by absorption law}\\
&\equiv [(p \wedge (\urcorner q \wedge \urcorner r)) \vee p] \wedge [(p \wedge \urcorner q \wedge \urcorner r) \vee (q \vee \urcorner r)]\\
&\equiv p \wedge [((p \wedge \urcorner q \wedge \urcorner r) \vee \urcorner r) \vee q], \text{ by absorption law}\\
&\equiv p \wedge (\urcorner r \vee q), \text{ by absorption law}\\
&\equiv p \wedge (q \vee \urcorner r)
\end{aligned}$$

PRINCIPAL DISJUNCTIVE AND PRINCIPAL CONJUNCTIVE NORMAL FORMS

The minterms

Given a number of variables, the products (or conjunctions) in which each variable or its negation, but not both, occurs only once are called the *minterms*.

1. Minterms for the two variables P and Q

$P \wedge Q, P \wedge \neg Q, \neg P \wedge Q, \neg P \wedge \neg Q.$

2. Minterms for the three variables P, Q and R

$P \wedge Q \wedge R, P \wedge Q \wedge \neg R, P \wedge \neg Q \wedge R, \neg P \wedge Q \wedge R, \neg P \wedge Q \wedge \neg R,$
$\neg P \wedge \neg Q \wedge R, P \wedge \neg Q \wedge \neg R, \neg P \wedge \neg Q \wedge \neg R$

Remark:

1. No minterms are equivalent.
2. Either $P \wedge Q$ or $Q \wedge P$ is included not both.

We note that there are 2^n minterms for n variables.

Principal disjunctive Normal form (PDNF):

For a given formula, an equivalent formula consisting of disjunctions of minterms only is known as its principal disjunctive normal form. Such a normal form is also called as the sum of products canonical form

Example 1. $P \to Q \Leftrightarrow (P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q)$

Remark: The number of minterms appearing in the PDNF is same as the number of entries with the truth value T in the truth table of $P \to Q$

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

Example 1. Obtain the principle disjunctive normal form of

$(P \to Q) \wedge (Q \to P)$

Solution:

$(P \to Q) \wedge (Q \to P) \Leftrightarrow (\neg P \vee Q) \wedge (\neg Q \vee P)$

$\Leftrightarrow ((\neg P \vee Q) \wedge \neg Q) \vee ((\neg P \vee Q) \wedge P)$

$\Leftrightarrow (\neg P \wedge \neg Q) \vee (Q \wedge \neg Q) \vee (P \wedge \neg P) \vee (Q \wedge P)$

$\Leftrightarrow (\neg P \wedge \neg Q) \vee F \vee F \vee (P \wedge Q)$

$\Leftrightarrow (P \wedge Q) \vee (\neg P \wedge \neg Q)$

Alternate Method: (Using truth table)

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Example 2. Obtain the principle disjunctive normal form of $P \rightarrow ((\mathbf{P} \rightarrow \mathbf{Q}) \wedge \neg(\neg Q \vee \neg P))$

Example 3. Obtain the principle disjunctive normal form of $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$

Solution:

$$(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$$

$$\Leftrightarrow (P \wedge Q \wedge T) \vee (\neg P \wedge R \wedge T) \vee (Q \wedge R \wedge T)$$

$$\Leftrightarrow (P \wedge Q \wedge (R \vee \neg R)) \vee (\neg P \wedge R \wedge (Q \vee \neg Q)) \vee (Q \wedge R \wedge (P \vee \neg P))$$

$$\Leftrightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge R \wedge Q) \vee (\neg P \wedge R \wedge \neg Q)$$

$$\vee (Q \wedge R \wedge P) \vee (Q \wedge R \wedge \neg P)$$

$$\Leftrightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (Q \wedge R \wedge \neg P) \vee (\neg P \wedge R \wedge \neg Q)$$

PRINCIPAL DISJUNCTIVE AND PRINCIPAL CONJUNCTIVE NORMAL FORMS

The maxterms

Given a number of variables, the sums (or disjunctions) in which each variable or its negation, but not both, occurs only once are called the *maxterms*.

Maxterms for the two variables P and Q

$P \vee Q, P \vee \neg Q, \neg P \vee Q, \neg P \vee \neg Q.$

Maxterms for three variables P, Q and R

$P \vee Q \vee R$	$P \vee \neg Q \vee R$	$\neg P \vee Q \vee \neg R$	$P \vee \neg Q \vee \neg R$
$P \vee Q \vee \neg R$	$\neg P \vee Q \vee R$	$\neg P \vee \neg Q \vee R$	$\neg P \vee \neg Q \vee \neg R$

Remark:

1. No max terms are equivalent.
2. Either $P \vee Q$ or $Q \vee P$ is included not both.
3. Maxterms are the dual of minterms.

Principal conjunctive Normal form (PCNF):

For a given formula, an equivalent formula consisting of conjunctions of maxterms only is known as its principal conjunctive normal form. Such a normal form is also called as the **product of sums** canonical form.

Example: $P \rightarrow Q \Leftrightarrow \neg P \vee Q$

Remark: The number of maxterms appearing in the PCNF is same as the number of entries with the truth value F in the truth table of $P \rightarrow Q$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Example:
Obtain the PCNF of $(\neg \mathbf{P} \rightarrow \mathbf{R}) \wedge (\boldsymbol{Q} \leftrightarrow \boldsymbol{P})$

Solution:

$(\neg \mathbf{P} \rightarrow \mathbf{R}) \wedge (\boldsymbol{Q} \leftrightarrow \boldsymbol{P})$

$\Leftrightarrow (\neg \neg \mathbf{P} \vee \mathbf{R}) \wedge ((\boldsymbol{Q} \longrightarrow \boldsymbol{P}) \wedge (\mathrm{P} \rightarrow \boldsymbol{Q}))$

$\Leftrightarrow (\mathbf{P} \vee \mathbf{R}) \wedge ((\neg \boldsymbol{Q} \vee \boldsymbol{P}) \wedge (\neg \mathrm{P} \vee \boldsymbol{Q}))$

$\Leftrightarrow (\mathbf{P} \vee \mathbf{R} \vee \boldsymbol{F}) \wedge ((\neg \boldsymbol{Q} \vee \boldsymbol{P}) \wedge (\neg \mathrm{P} \vee \boldsymbol{Q}))$

$\Leftrightarrow (\mathbf{P} \vee \mathbf{R} \vee \boldsymbol{F}) \wedge (\neg \boldsymbol{Q} \vee \boldsymbol{P} \vee \boldsymbol{F}) \wedge (\neg \mathrm{P} \vee \boldsymbol{Q} \vee \boldsymbol{F})$

$\Leftrightarrow (\mathbf{P} \vee \mathbf{R} \vee (\mathbf{Q} \wedge \neg \mathbf{Q})) \wedge (\neg \mathbf{Q} \vee \mathbf{P} \vee (\mathbf{R} \wedge \neg \mathbf{R})) \wedge (\neg \mathrm{P} \vee \mathbf{Q} \vee (\mathbf{R} \wedge \neg \boldsymbol{R}))$

$\Leftrightarrow (\mathbf{P} \vee \mathbf{R} \vee \mathbf{Q}) \wedge (\mathbf{P} \vee \neg \mathbf{Q} \vee \mathbf{R}) \wedge (\mathrm{P} \vee \neg \mathbf{Q} \vee \boldsymbol{R}) \wedge (\mathbf{P} \vee \neg \mathbf{Q} \vee \neg \mathbf{R})$

$\wedge (\neg \mathrm{P} \vee \mathbf{Q} \vee \boldsymbol{R}) \wedge (\neg \mathrm{P} \vee \mathbf{Q} \vee \neg \boldsymbol{R})$

$\Leftrightarrow (\mathbf{P} \vee \mathbf{R} \vee \mathbf{Q}) \wedge (\mathbf{P} \vee \neg \mathbf{Q} \vee \mathbf{R}) \wedge (\mathbf{P} \vee \neg \mathbf{Q} \vee \neg \mathbf{R}) \wedge (\neg \mathrm{P} \vee \mathbf{Q} \vee \mathbf{R}) \wedge (\neg \mathrm{P} \vee \mathbf{Q} \vee$

$\neg \mathbf{R})$

PCNF through PDNF

Example: Obtain the principle conjunctive normal form of $(P \wedge Q) \vee (\neg P \wedge Q \wedge R)$

Solution:

$$A = (P \wedge Q) \vee (\neg P \wedge Q \wedge R)$$

$$\Leftrightarrow (P \wedge Q \wedge T) \vee (\neg P \wedge Q \wedge R)$$

$$\Leftrightarrow (P \wedge Q \wedge (R \vee \neg R)) \vee (\neg P \wedge Q \wedge R)$$

$$\Leftrightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R)$$

This is PDNF of the given formula.

$$A = (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R)$$

$\neg A$ contains the remaining terms

$$\neg \boldsymbol{A} = (\neg \boldsymbol{P} \wedge \neg \boldsymbol{Q} \wedge \neg \boldsymbol{R}) \vee (\neg \boldsymbol{P} \wedge \neg \boldsymbol{Q} \wedge \boldsymbol{R}) \vee (\neg \boldsymbol{P} \wedge \boldsymbol{Q} \wedge \neg \boldsymbol{R}) \vee (\boldsymbol{P} \wedge \neg \boldsymbol{Q} \wedge \boldsymbol{R}) \vee (\boldsymbol{P} \wedge \neg \boldsymbol{Q} \wedge \neg \boldsymbol{R})$$

We know that $\neg \neg A = A$. Taking negation of $\neg A$, we get PCNF

$$\neg \neg \mathbf{A}$$

$$\Leftrightarrow \neg (\neg \mathbf{P} \wedge \neg \mathbf{Q} \wedge \neg \mathbf{R}) \wedge \neg (\neg \mathbf{P} \wedge \neg \mathbf{Q} \wedge \mathbf{R}) \wedge \neg (\neg \mathbf{P} \wedge \mathbf{Q} \wedge \neg \mathbf{R}) \wedge \neg (\mathbf{P} \wedge \neg \mathbf{Q} \wedge \mathbf{R}) \wedge \neg (\mathbf{P} \wedge \neg \mathbf{Q} \wedge \neg \mathbf{R})$$

$$\Leftrightarrow (\mathbf{P} \vee \mathbf{Q} \vee \mathbf{R}) \wedge (\mathbf{P} \vee \mathbf{Q} \vee \neg \mathbf{R}) \wedge (\mathbf{P} \vee \neg \mathbf{Q} \vee \mathbf{R}) \wedge (\neg \mathbf{P} \vee \mathbf{Q} \vee \neg \mathbf{R}) \wedge (\neg \mathbf{P} \vee \mathbf{Q} \vee \mathbf{R})$$

is the PCNF of A.

Inference Theory for Statement Calculus

- The main function of logic is to provide rules for inference.
- The rules for inference are criteria for determining the validity of an argument.
- Inference theory is concerned with the inferring of a conclusion from certain hypotheses or basic assumptions, called **premises**, by applying certain principles of reasoning, called **rules of inference**.
- When a conclusion derived from a set of premises by using rules of inference, the process of such derivation is called a **deduction** or **formal proof**.
- Any conclusion which is arrived at by following these rules is called a **valid conclusion** and the argument is called a **valid argument**.

Theory of Inference:

Let A and B be two statement formulas. Then "B logically follows from A" or " B is a valid conclusion (consequence) of the premise A" if and only if $A \to B$ is a tautology and is denoted by $A \Rightarrow B$.

An argument is a sequence of propositions $H_1, H_2, \ldots, H_n$ called premises (or hypotheses) followed by a proposition C called conclusion. An argument is usually written:

$$\begin{array}{l} H_1 \\ H_2 \\ \vdots \\ H_n \\ \text{implies} \quad C \end{array}$$

or

$$H_1 \wedge H_2 \wedge \ldots \wedge H_n \Rightarrow C$$

If a set of premises and a conclusion are given, it is possible to determine whether the conclusion follows from the premises by constructing relevant truth tables, as explained in the following example. This method is known as truth table technique.

Example: $H_1 : p$ and $H_2 : p \rightarrow q$ then $C : q$ (Modus Ponens)

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Notice that $(p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology. Therefore it is valid.

Thus $p,\ p \rightarrow q \Rightarrow q$.

Example: $H_1 : q$ and $H_2 : p \to q$ then $C : p$

p	q	$p \to q$	$q \wedge (p \to q)$	$(q \wedge (p \to q)) \to p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

Notice that $q \wedge (p \to q) \to p$ is a contigency. Therefore it is invalid.

Example 1.1 Find whether the conclusion C follows from the premises H_1, H_2, H_3 in the following cases, using truth table technique:

(i) H_1: $\neg p$, H_2: $p \vee q$, C: $p \wedge q$

(ii) H_1: $p \vee q$, H_2: $p \to r$, H_3: $q \to r$, C: r

IMPLICATION TABLE:

S.No	Formula	Name
1	$p \wedge q \Rightarrow p$	simplification
	$p \wedge q \Rightarrow q$	
2	$p \Rightarrow p \vee q$	addition
	$q \Rightarrow p \vee q$	
3	$p, \; q \Rightarrow p \wedge q$	Conjunction
4	$p, \; p \rightarrow q \Rightarrow q$	modus ponens
5	$\neg p, \; p \vee q \Rightarrow q$	disjunctive syllogism
6	$\neg q, \; p \rightarrow q \Rightarrow \neg p$	modus tollens
7	$p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$	Hypothetical syllogism

Rules of inference: (templates for constructing valid argument)

Rule P: A premises can be introduced at any step of derivation.

Rule T: A formula can be introduced provided it is Tautologically implied by previously introduced formulas in the derivation.

Example: Show that $\neg q$ and $p \to q$ implies $\neg p$.

Solution: (A formal proof is as follows):

Step 1. $p \to q$ Rule P

Step 2. $\neg q \to \neg p$ Rule T

Step 3. $\neg q$ Rule P

Step 4. $\neg p$ Combined $\{2, 3\}$ and apply Modus Ponens

Example: Show that r is a valid inference from the premises $p \rightarrow q$, $q \rightarrow r$ and p.

Solution:

Step	Premises	Rule
1	p	Rule P
2	$p \rightarrow q$	Rule P
3	q	Rule T, {1,2} Modus Ponens
4	$q \rightarrow r$	Rule P
5	r	Rule T {3,4} Modus Ponens

Example: Prove that S is a valid conclusion from the premises $P \to \neg Q$, $Q \vee R$, $\neg S \to P$ and $\neg R$.

Answer:

Step	Statement formula	Rule
1.	$\neg R$	P
2.	$Q \vee R$	P
3.	Q	T, [(1), (2), Disjunctive Syllogism]
4.	$P \to \neg Q$	P
5.	$\neg P$	T, [(3), (4), Modus Tollens]
6.	$\neg S \to P$	P
7.	$\neg(\neg S)$	T [(5), (6), Modus Tollens]
8	S	T [(7), Double negation

Example: Consider the following statements: 'I take the bus or I walk. If I walk I get tired. I do not get tired. Therefore I take the bus.'

Solution:

We can formalize this by calling B= I take the bus, W= I walk and J= I get tired. The premises are $B \vee W$, $W \rightarrow J$ and $\neg J$, and the conclusion is B. The argument can be described in the following steps:

step	statement	reason
1	$W \rightarrow J$	P
2	$\neg J$	P
3	$\neg W$	$\{1, 2\}$, Modus Tollens
4	$B \vee W$	P
5	B	$\{3, 4\}$, Disjunctive Syllogism

Rule CP: If the conclusion is of the form $r \to s$ then we include r as an additional premises and derive s. CP: conditional proof

Indirect method: We use negation of the conclusion as an additional premise and try to arrive a contradiction.

Inconsistent: A set of premises are inconsistent provided their conjunction implies a contradiction.

Show that $r \to s$ can be derived from the premises $p \to (q \to s)$, $\neg r \vee p$ and q.

* Instead of deriving $r \to s$, an addition premise r can be included and arrive at s.

Step No.	Statement formula	Rule
1	$\neg r \vee p$	Rule P
2	$r \to p$	Rule T, (1): $p \to q \Leftrightarrow \neg p \vee q$
3	r	Rule P (Assumed premise)
4	p	Rule T, (2),(3); p, $p \to q \Rightarrow q$ (modus ponens)
5	$p \to (q \to s)$	Rule P
6	$q \to s$	Rule T, (4),(5). p, $p \to q \Rightarrow q$
7	q	Rule P
8.	s	Rule T (6) (7) modus Ponen
9	$r \to s$	Rule CP

Example: Prove that the premises $\neg q$, $p \to q$ result in the conclusion $\neg p$ by the indirect method of proof.

Answer:

We now include $\neg\neg p$ as an additional premise. The argument form is given below:

Step	Statement formula	Rule
1.	$\neg\neg p$	P (additional premise)
2.	p	T [1, double negation]
3.	$p \to q$	P
4.	$\neg q \to \neg p$	T [3, contrapositive]
5.	$\neg q$	P
6.	$\neg p$	T [4, 5, Modus ponens]
7.	$p \wedge \neg p$	T [2, 6, conjunction]

Thus the inclusion of $\neg C$ leads to a contradiction. Hence $\neg q$, $p \to q \Rightarrow \neg p$

Example: Prove that $p \to q$; $p \to r$; $q \to \neg r$ and p are inconsistent.

Answer: The desired result is false.

Step	Statement formula	Rule
1.	p	P
2.	$p \to q$	P
3.	q	T [1, 2, Modus Ponens]
4.	$q \to \neg r$	P
5.	$\neg r$	T [3, 4, Modus Ponens]
6.	$p \to r$	P
7.	$\neg p$	T [5, 6, Modus Tollens]
8.	F	T [1, 7, conjunction]

Predicate Calculus

Consider the statement

- John is a bachelor
- Smith is a bachelor

To reveal common features of these statements

We denote

Predicate part <u>is a bachelor</u> as <u>P</u>

Subjects: John as <u>j</u> and Smith as <u>s</u>

Then above statements in predicate form are

- $P(j)$: John is a bachelor
- $P(s)$: Smith is a bachelor

In this section we extend the system of logic to include such an above statements.

Predicate: A part of a declarative sentence describing the properties of an object or relation among the objects.

Ex: Consider the statement "John is a player".

Here `is a player' is called a predicate

and

`John' is a subject.

This is an example of one place predicate.

Type of Predicates:

- One place predicate:

  This painting is red.

  Can be symbolized as $R(p)$.

- Two place predicate:

  Ex: Mani is taller than Ravi.

  We can denote this as $T(m,r)$.

T is two place predicate as two subjects needed to complete the statement.

3-place predicate:

Ex: Mohan stands between Raj and Nitin.

B(m,r,n)

4-place predicate:

David & Tendulkar played cricket against Ganguly and Kambli.

P(d,t,g,k)

Similarly if there are n variables in a predicate, then it is called n-ary predicate or n-place predicate.

The truth value of p(x) can be predicted on the values assigned to x from the universe of discourse (UD) (the set of values that x can take).

Compound statements:

It is obtained by combining one or more simple statements by logical connectives.

- Example:

John is bachelor.

This painting is red.

If we denote these statement as $B(j)$ and $R(p)$

Then compound statements are:

$B(j) \wedge R(p)$, $\sim B(j)$, $B(j) \rightarrow R(p)$ etc.

Statement Function of one variable:

A simple statement function of one variable is defined to be an expression consisting of a predicate symbol and an individual variable.

Such a statement function becomes a statement when the variable is replaced by the name of the object.

Ex: $H(x)$: x is mortal.

If we denote Ram by r, Mohan by m, Canada by c.

Then $H(r), H(m), H(c)$ all denote statements.

Compound Statement Functions:

Ex: Let $M(x)$: x is a man.

$H(x)$: x is mortal.

Then compound statement functions are

$M(x) \wedge H(x)$, $\sim H(x)$, $H(x) \rightarrow M(x)$, $M(x) \vee \sim H(x)$ $etc.$

Same way we can define a statement function of two variables and more:

Ex: Let $G(x, y)$: x is taller than y.

If x is replaced by Pooja and y by Megha.

Then $G(x, y)$ becomes

Pooja is taller than Megha.

It is possible to form statement function of two variable from statement function of one variable:

Ex: M(x): x is a man.

H(y): y is mortal.

Then $M(x) \wedge H(y)$: x is a man and y is mortal.

Quantifiers:

We can indicate the quantity by means of word like all, some, none etc. Such words in the statement are called quantifier.

Ex: Some men are mortal.

All flower contains nectar.

There are two type of quantifiers:-

1. Universal quantifier
2. Existential quantifier

Universal Quantifier:

The quantifier all is called universal quantifier. It is represented by the symbol (∀x) or (x) for variable x.

(∀x) or (x) represents each of the phrases:

1. For all x
2. For every x
3. For each x
4. Everything x such that
5. Each thing x is such that

Universal Quantifier:

Consider the statement

(1) All odd primes are greater than 2.

The word all in this preposition is a logical quantifier.

This can be written as
For every x, if x is an odd prime then x is greater than 2.
i.e
$O(x)$: x is a odd prime.
$G(x)$: x is greater than 2.

Then symbolic form of above statement is

$$(\forall x)\ (O(x) \rightarrow G(x))$$

Universal Quantifier:

Consider the statement

(2) All men are mortal.

This can be written as ,
For every x, If x is a men then x is mortal.

Let us denote

$M(x)$: x is man.
$T(x)$: x is mortal.

Then symbolic form of above statement is

$$(\forall x)\ (M(x) \rightarrow T(x))$$

Every rational number is a real number may be translated as.
For every x, if x is a rational number, then x is a real number.

The symbolic form of above statement is

$$(\forall x)\ (R(x) \rightarrow N(x))$$

- $(\forall x)M(x)$ can be translated as

  For all x, x is a man.

  or

  For every x, x is a man.

- Let domain is the set S={x_1,x_2,....., x_n}

Then $(\forall x)P(x)$=$P(x_1) \wedge P(x_2) \wedge \ldots\ldots \wedge P(x_n)$

Existential Quantifier:

The quantifier Some is called existential quantifier. It is represented by the symbol $(\exists x)$.

$(\exists x)$ represents each of the phrases:

1. For some x
2. Some x such that
3. There is an x such that
4. There exist x such that
5. There is at least one x such that.

Existential Quantifier:

Ex: Some apples are red.

A(x): x is a apple.
R(x): x is red.

This statement can be written as
There exist x such that x is a apple and x is red.

i.e symbolic form of above statement is

$(\exists x)$ (A$(x) \wedge R(x))$

- Let domain is the set S=$\{x_1, x_2, \ldots\ldots, x_n\}$

Then $(\exists x)P(x)$=$P(x_1) \vee P(x_2) \vee \ldots\ldots \vee P(x_n)$

Note-1: in symbolic expressions of the type "All A are B" the correct connective that should be used is conditional.

Note-2: in symbolic expressions of the type "Some A are B" the correct connective that should be used is conjunction.

Free and Bound Variable

A part of the formula containing $(\forall x)P(x)$ or $(\exists x)\ P(x)$ is called x-bound part of the formula.

An occurrence of x in x bound part of a formula is called bound occurrence.

Any occurrence of x that is not bound occurrence if free occurrence.

Scope of the quantifier

The formula $P(x)$ in $(\forall x)P(x)$ or in $(\exists x)\ P(x)$ called scope of the quantifier i.e. scope of the quantifier is the formula immediately following the quantifier.

Ex: $(\forall x)P(x, y)$
Here $P(x, y)$ is the scope of the quantifier.
Both occurrences of x is bound occurrences while occurrence of y is free occurrence.

Ex:$(\forall x)(P(x) \to Q(x))$
Scope of the quantifier is $P(x) \to Q(x)$ and all occurrences of are bound occurrences.

Ex::$(\forall x)(P(x)) \to Q(x)$

Ex 1. Find the truth value of $(\forall x)P(x)$, where $P(x){:}x^2 < 10$ and universe of discourse consist of a positive integer not exceeding 4.

Solution: The statement $(\forall x)P(x)$ for the given universe of discourse, is same as

$$P(1) \wedge P(2) \wedge P(3) \wedge P(4) \Leftrightarrow T \wedge T \wedge T \wedge F$$
$$\Leftrightarrow \text{F}$$

i.e. $(\forall x)P(x)$ is false.

Ex 2. Find the truth value of $(\forall x)\big(P \to Q(x)\big) \vee R(a)$, where

$P{:}\, 2 > 1,\ Q(x){:}\, x \leq 3,\ R(x){:}\, x > 5\ and\ a = 5$, the universe of discourse is $\{-2,3,6\}$.

Ans: F

- Determine the truth value if A={1,2,3,……,9,10}

(a) $(x)(\exists y)(x + y < 14)$ Ans: True

(b) $(x)(y)(x + y < 14)$ Ans: False

Equivalence Formula of Predicate Calculus:

1. $(\exists x)\big(\mathrm{A}(\mathrm{x}) \vee B(x)\big) \Leftrightarrow (\exists x)\mathrm{A}(\mathrm{x}) \vee (\exists x)\mathrm{B}(\mathrm{x})$
2. $(\forall\, x)(A(x) \wedge B(x)) \Leftrightarrow (\forall\, x)A(x) \wedge (\forall\, x)B(x)$
3. $(\forall\, x)(A \vee B(x)) \Leftrightarrow A \vee (\forall\, x)B(x)$
4. $(\exists x)(A \wedge B(x)) \Leftrightarrow A \wedge (\exists x)B(x)$
5. $(\forall\, x)A(x) \rightarrow B \Leftrightarrow (\exists x)(A(x) \rightarrow B)$
6. $(\exists x)A(x) \rightarrow B \Leftrightarrow (\forall x)(A(x) \rightarrow B)$
7. $A \rightarrow (\forall\, x)B(x) \Leftrightarrow (\forall x)\big(A \rightarrow B(x)\big)$
8. $A \rightarrow (\exists x)B(x) \Leftrightarrow (\exists x)\big(A \rightarrow B(x)\big)$

Implications of Predicate Calculus

1. $(\forall x)\mathrm{A}(\mathrm{x}) \vee (\forall x)B(x) \Rightarrow (\forall x)\big(\mathrm{A}(\mathrm{x}) \vee \mathrm{B}(\mathrm{x})\big)$

2. $(\exists x)\big(\mathrm{A}(\mathrm{x}) \wedge B(x)\big) \Rightarrow (\exists x)\mathrm{A}(\mathrm{x}) \wedge (\exists x)\mathrm{B}(\mathrm{x})$

Demorgan's Laws for Predicate Calculus:

1. $\urcorner(\forall x)P(x) \Leftrightarrow (\exists x)\urcorner P(x)$
2. $\urcorner(\exists x)P(x) \Leftrightarrow (\forall x)\urcorner P(x)$

Proof 1: Let domain is the set S={x_1,x_2,....., x_n}

Then $(\forall x)P(x)$=$P(x_1)\wedge P(x_2)\wedge \ldots\ldots \wedge P(x_n)$

And $(\exists x)P(x)$=$P(x_1)\vee P(x_2)\vee \ldots\ldots \vee P(x_n)$

Now, $\urcorner(\forall x)P(x) \Leftrightarrow \urcorner\,[P(x_1)\wedge P(x_2)\wedge \ldots\ldots \wedge P(x_n)]$

$$\Leftrightarrow \urcorner P(x_1)\vee \urcorner P(x_2)\vee \ldots\ldots \vee \urcorner P(x_n)$$

$$\Leftrightarrow (\exists x)\urcorner P(x)$$

i.e. $\urcorner(\forall x)P(x) \Leftrightarrow (\exists x)\urcorner P(x)$

TRANSLATING FROM ENGLISH TO LOGIC

Example 1: Translate the following sentence into predicate logic: "Every student in this class has taken a course in Java."

Solution:

First decide on the domain U.

Solution 1: If U is all students in this class, define a propositional function J(x) denoting "x has taken a course in Java" and translate as $\forall x\, J(x)$.

Solution 2: But if U is all people, also define a propositional function S(x) denoting "x is a student in this class" and translate as $\forall x\, (S(x) \rightarrow J(x))$.

Example 2: Translate the following sentence into predicate logic: "Some student in this class has taken a course in Java."

Solution:

First decide on the domain U.

Solution 1: If U is all students in this class, translate as

$$\exists x\, J(x)$$

Solution 2: But if U is all people, then translate as $\exists x\, (S(x) \land J(x))$

NEGATING QUANTIFIED EXPRESSIONS

- Consider $\forall x\, J(x)$

  "Every student in your class has taken a course in Java."

  Here $J(x)$ is "x has taken a course in Java" and

  the domain is students in your class.

- Negating the original statement gives "It is not the case that every student in your class has taken Java." This implies that "There is a student in your class who has not taken Java."

  Symbolically $\neg\forall x\, J(x)$ and $\exists x\, \neg J(x)$ are equivalent

- Now Consider $\exists\, x\, J(x)$

  "There is a student in this class who has taken a course in Java."

  Where $J(x)$ is "x has taken a course in Java."

- Negating the original statement gives "It is not the case that there is a student in this class who has taken Java." This implies that "Every student in this class has not taken Java"

  Symbolically $\neg\exists\, x\, J(x)$ and $\forall\, x\, \neg J(x)$ are equivalent

In the negation a proposition 'for all' becomes 'there is' and 'there is' becomes 'for all' i.e., the symbol $\forall$ becomes $\exists$ and $\exists$ becomes $\forall$.

NESTED (MORE THAN ONE) QUANTIFIERS

Definition: Two quantifiers are said to be nested if one is within the scope of other.

Note: Anything within the scope of the quantifier can be thought as propositional function

$$(\forall x)\underbrace{\boxed{(\exists y)Q(x,y)}}_{P(x)} \Rightarrow (\forall x)P(x)$$

Note: In the nested case, if both the quantifiers are same, then ordering does not matter.

- **Determine the truth value if $A=\{1,2,3,\ldots\ldots,9,10\}$**

(a) $(\forall x)(\exists y)(x+y<14)$ Ans: True

(b) $(\forall x)(\forall y)(x+y<14)$ Ans: False

TRANSLATING NESTED QUANTIFIERS INTO ENGLISH

Example: Translate the statement

$$\forall x \ (C(x) \lor \exists y \ (C(y) \land F(x, y)))$$

where C(x) is "*x* has a computer," and *F*(*x*,*y*) is "*x* and *y* are friends," and the domain for both *x* and *y* consists of all students in your school.

Solution: Every student in your school has a computer or has a friend who has a computer.

QUESTIONS ON TRANSLATION FROM ENGLISH

Choose the obvious predicates and express in predicate logic.

Example 1: "Brothers are siblings."

Solution: $\forall x\ \forall y\ (B(x,y) \rightarrow S(x,y))$

Example 2: "Everybody loves somebody."

Solution: $\forall x\ \exists y\ L(x,y)$

Example 3: "There is someone who is loved by everyone."

Solution: $\exists y\ \forall x\ L(x,y)$

Example 4: "There is someone who loves someone."

Solution: $\exists x\ \exists y\ L(x,y)$

Example 5: "Everyone loves himself"

Solution: $\forall x\ L(x,x)$

Inference Theory for Predicate Calculus:

Let A(x) be any predicate formula, where x is particular object.

1. Universal Specification (Rule U.S) or Universal Instantiation: It is the rule of inference which states that one can conclude that $A(y)$ is true, if $(\forall x)\ A(x)$ is true, where y is any arbitrary member of the universe of discourse.

   i.e. $$(\forall x)\ A(x) \Rightarrow A(y)$$

   If y is variable then it must not be quantified somewhere in A(x)

2. Universal Generalization (Rule U.G): It is the rule which states that $(\forall x)\ A(x)$ is true, if $A(y)$ is true, where y is the arbitrary member of the universe of discourse.

   i.e. $$A(y) \Rightarrow (\forall x)\ A(x)$$

   x must not appear as free in A(y)

3. Existential Specification (Rule E.S) : Existential Specification is the rule of inference which allows us to conclude that $A(y)$ is true, if $(\exists x)\ A(x)$ is true where y is not an arbitrary member of the universe, but one for which $A(y)$ is true.

   i.e. $$(\exists x)\ A(x) \Rightarrow A(y)$$

   y must be a new symbol.

4. Existential Generalization (Rule E.G): Existential Generalization is the rule that is used to conclude that $(\exists x)\ A(x)$ is true, when $A(y)$ is true, where y is a particular member of the universe of discourse.

   $$A(y) \Rightarrow (\exists x)\ A(x)$$

   x must not appear free in A(y)

(i) Show that $(x)(H(x) \longrightarrow M(x)) \wedge H(a) \Longrightarrow M(a)$.

Solution:

Step 1	$(x)(H(x) \longrightarrow M(x))$	Rule P
Step 2	$H(a) \longrightarrow M(a)$	Rule US
Step 3	$H(a)$	Rule P
Step 4	$M(a)$	$\{2, 3\}$ and apply Modus Ponens

xample:

$$(x)(P(x) \longrightarrow Q(x)) \wedge (x)(Q(x) \longrightarrow R(x)) \Longrightarrow (x)(P(x) \longrightarrow R(x)).$$

Solution:

Step 1	$(x)(P(x) \longrightarrow Q(x))$	Rule P
Step 2	$P(a) \longrightarrow Q(a)$	Rule US
Step 3	$(x)(Q(x) \longrightarrow R(x))$	Rule P
Step 4	$Q(a) \longrightarrow R(a)$	Rule US
Step 5	$P(a) \longrightarrow R(a)$	$\{2,4\}$, **Rule T**
Step 6	$(x)(P(x) \longrightarrow R(x))$.	Rule UG

Example:

Show that $(\exists x)(P(x) \wedge Q(x)) \implies (\exists x)P(x) \wedge (\exists x)Q(x)$

Solution:

Step 1	$(\exists x)(P(x) \wedge Q(x))$	Rule P
Step 2	$P(a) \wedge Q(a)$	Rule ES
Step 3	$P(a)$	Rule T
Step 4	$Q(a)$	Rule T
Step 5	$(\exists x)P(x)$	{3},EG
Step 6	$(\exists x)Q(x)$	{4},EG
Step 7	$(\exists x)P(x) \wedge (\exists x)Q(x)$	{5,6}, Rule T

xample: Show that from

(a) $(\exists x)(F(x) \wedge S(x)) \longrightarrow (y)(M(y) \longrightarrow W(y))$

(b) $(\exists y)(M(y) \wedge \neg W(y))$

the conclusion is $(x)(F(x) \longrightarrow \neg S(x))$.

Solution:

Step 1	$(\exists y)(M(y) \wedge \neg W(y))$	Rule P
Step 2	$(M(a) \wedge \neg W(a))$	ES
Step 3	$\neg(M(a) \longrightarrow W(a))$	Rule T
Step 4	$(\exists y)\neg(M(y) \longrightarrow W(y))$	EG
Step 5	$\neg(y)(M(y) \longrightarrow W(y))$	Rule T
Step 6	$(\exists x)(F(x) \wedge S(x)) \longrightarrow (y)(M(y) \longrightarrow W(y))$	Rule P
Step 7	$\neg(\exists x)(F(x) \wedge S(x))$	Rule T{5,6}
Step 8	$(x)\neg(F(x) \wedge S(x))$	Rule T
Step 9	$\neg(F(a) \wedge S(a))$	US
Step 10	$F(a) \longrightarrow \neg S(a)$	Rule T
Step 11	$(x)(F(x) \longrightarrow \neg S(x))$	UG

Indirect Method of Proof:

Example: Show that $(x)(P(x) \vee Q(x)) \Longrightarrow (x)P(x) \vee (\exists x)Q(x)$ by indirect method

Solution: Proof by indirect method

Step 1	$\neg((x)P(x) \vee (\exists x)Q(x))$	Rule P
Step 2	$\neg(x)P(x) \wedge \neg(\exists x)Q(x)$	Rule T
Step 3	$\neg(x)P(x)$	Rule T
Step 4	$\neg(\exists x)Q(x)$	Rule T
Step 5	$(\exists x)\neg P(x)$	3,Rule T
Step 6	$(x)\neg Q(x)$	4,Rule T
Step 7	$\neg P(a)$	5,ES
Step 8	$\neg Q(a)$	6,US
Step 9	$\neg P(a) \wedge \neg Q(a)$	Rule T {7,8}
Step 10	$\neg(P(a) \vee Q(a))$	Rule T
Step 11	$(x)(P(x) \vee Q(x))$	Rule P
Step 12	$P(a) \vee Q(a)$	US
Step 13	$\neg(P(a) \vee Q(a)) \wedge (P(a) \vee Q(a))$	Rule T {10,12}
Step 14	F	Rule T

Example: **Use conditional proof to prove that**

$$(x)(P(x) \rightarrow Q(x)) \Rightarrow (x)P(x) \rightarrow (x)Q(x)$$

Solution:

Take $(x)P(x)$ as an assumed premise

1.	$(x)(P(x) \rightarrow Q(x))$	Rule P
2.	$P(y) \rightarrow Q(y)$	Rule US, (1)
3.	$(x)P(x)$	Rule P [Assumed premise]
4.	$P(y)$	Rule US, (3)
5.	$Q(y)$	Rule T, (2), (4) $P, P \rightarrow Q \Rightarrow Q$
6.	$(x)Q(x)$	Rule UG, (5)
7.	$(x)P(x) \rightarrow (x)Q(x)$	Rule C.P

Hence proved.